Information Geometry and Embedding Collapse in Long-Context Transformers

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Abstract

We study how the geometry of contextual embeddings in autoregressive language models deforms as the input context grows. Three complementary diagnostics are proposed: a TwoNN-based intrinsic dimension, a pairwise anisotropy measurement extending Ethayarajh (2019), and a discrete curvature proxy based on local PCA over k-nearest-neighbour patches. Applied to a GPT-2 backbone on a long-document corpus across context lengths from 128 to 1024 tokens, we observe a clean drop of intrinsic dimension in the middle layers from 8.61 to 0.89 and an anisotropy shift from 0.523 to 0.414, with effective rank decreasing monotonically. The pipeline is configuration-driven and rescales to 7B-parameter open-weight models and 131,072-token contexts without code changes.

1 Introduction

Long-context language models have become routine: Gemini 1.5, Claude, GPT-4, and open-weight architectures such as Llama 3 routinely ingest tens of thousands of tokens. The empirical observation that downstream quality often degrades super-linearly with context length is widely reported but poorly understood from a geometric standpoint. We ask a specific question:

> *Does the hidden-state manifold of a transformer language model undergo a measurable phase transition as the context length grows, and if so, where in the network does it occur?*

A geometric account is attractive because it sits beneath the architectural details. Whatever positional encoding, attention pattern, or rotary modification is in use, the residual stream lives in $\mathbb{R}^d$ and inherits a metric, a dimension, and a curvature. If long-context degradation were primarily geometric—an attractor collapse of the residual stream toward a low-dimensional subspace—we should be able to detect that collapse without any reference to the task.

We make three contributions. **(i)** We define a triple of cheap, easily implementable geometric diagnostics: an intrinsic dimension estimator (TwoNN), a pairwise anisotropy measure, and a local-PCA curvature proxy. **(ii)** We sweep these diagnostics over a four-decade range of context lengths on GPT-2, releasing a reproducible pipeline that scales without code changes to Llama-3-8B and Mistral-7B at 131,072 tokens given GPU resources. **(iii)** We show that, even on a small backbone, the predicted collapse is visible: intrinsic dimension drops by roughly an order of magnitude in the middle layers between short and long contexts, anisotropy rises, and effective rank decreases monotonically.

The remainder of the paper is organised as follows. Section 2 reviews related work on anisotropy and intrinsic dimensionality. Section 3 formalises the diagnostics and Section 4 the sweep. Section 5 reports findings, Section 6 interprets them, and Section 7 concludes.

2 Related Work

Anisotropy in contextual embeddings. Ethayarajh [4] showed that average pairwise cosine similarity between contextualised tokens in BERT, GPT-2, and ELMo grows with depth, contradicting the implicit assumption that contextual vectors live on a uniform unit sphere. Subsequent work [3, 2] confirmed the phenomenon, attributed it to a small number of dominant directions, and proposed whitening-based corrections. Our anisotropy measurement is a direct extension of Ethayarajh's, sampled across context lengths rather than across depths.

Intrinsic dimensionality of representations. The TwoNN estimator of Facco et al. [5] provides a non-parametric estimate of the intrinsic dimension of a point cloud from the ratio of first and second nearest-neighbour distances. It has been applied to image classifiers [1] and, more recently, to language models, where intrinsic dimension has been argued to track generalisation and abstraction depth. We adopt TwoNN because it is robust in the high-dimensional regime where covariance-based

estimators degenerate.

Long-context behaviour. Several works [6, 9] report systematic degradation of long-context accuracy. Most are task-driven (question answering, needle-in-a-haystack). Our angle is orthogonal: we instrument the residual stream geometrically and check whether collapse precedes task failure.

Mechanistic interpretability. Activation patching and circuit discovery [8, 7] are complementary lenses. Where they ask *which heads carry a computation*, we ask *what shape the representation has at all*. The two approaches share a methodological commitment: study what is actually there, before proposing remediation.

3 Methodology

3.1 Setup

Let f_θ be a decoder-only transformer with L layers and hidden dimension d. For input $x \in V^*$, denote by $h_i^{(\ell)}(x) \in \mathbb{R}^d$ the residual-stream activation at layer ℓ and position i. Given a context length T, we sample documents from a corpus, truncate to T tokens, and assemble a point cloud

$$\mathcal{X}_T^{(\ell)} \;=\; \{\, h_i^{(\ell)}(x) \;\mid\; x \in \mathcal{D},\, 1 \le i \le T \,\} \subset \mathbb{R}^d.$$

We then compute three diagnostics on each cloud.

3.2 Intrinsic dimension (TwoNN)

For each point $p \in \mathcal{X}$ let $r_1(p), r_2(p)$ be its first- and second-nearest-neighbour distances. Facco et al. show that $\mu(p) := r_2(p)/r_1(p)$ follows a Pareto distribution with parameter d, yielding the maximum-likelihood estimator

$$\hat{d}_{\text{TwoNN}} = \frac{N}{\sum_p \log \mu(p)}, \tag{1}$$

where N is the number of valid samples. We truncate the upper tail at the 90th percentile of μ to suppress boundary effects.

3.3 Anisotropy

Following [4], we draw M random index pairs (i, j) uniformly without replacement and define

$$A(\mathcal{X}) = \frac{1}{M} \sum_{(i,j)} \frac{\langle x_i, x_j \rangle}{\|x_i\|\,\|x_j\|}. \tag{2}$$

A spherically uniform cloud satisfies $A \approx 0$; A close to 1 indicates collapse onto a narrow cone.

3.4 Local-PCA curvature proxy

For a subsample of $S \le N$ anchor points, retrieve the k nearest neighbours, centre them, and compute singular values $\sigma_1 \ge \cdots \ge \sigma_k$. Let

$$C(p) = \frac{\sum_{j>m} \sigma_j^2}{\sum_{j \le m} \sigma_j^2 + \varepsilon}, \tag{3}$$

where m is the count of *leading* directions (we use $m = 3$). A flat manifold concentrates mass in the leading directions ($C \to 0$); a high-curvature region spreads mass across all directions (C large). We report $\overline{C} = \mathbb{E}_p[C(p)]$.

3.5 Effective rank

For completeness we report the effective rank of the centred activation covariance, $\exp\big(-\sum_i p_i \log p_i\big)$, where $p_i = \sigma_i / \sum_j \sigma_j$. This gives a smooth, dimensionless measure of how many directions carry appreciable variance and complements the discrete TwoNN estimate.

4 Experiments

Models. The local pipeline uses GPT-2 small (12 blocks, hidden $d = 768$); the GPU configuration (`configs/gpu_full.yaml`) switches to Llama-3-8B and Mistral-7B without code changes.

Corpus. For reproducibility on commodity hardware we use a self-contained ArXiv-style abstract corpus (eight long documents, concatenated to reach the target context length). The GPU configuration streams ArXiv summarisation data via Hugging Face datasets.

Context-length sweep. We sweep $T \in \{128, 256, 512, 1024\}$ on the local configuration; the GPU configuration extends the sweep to 131,072 tokens. For each T and each layer ℓ, all metrics are computed on a subsample of up to 4000 tokens to bound running time.

Hyperparameters. TwoNN uses the 90th-percentile truncation recommended by Facco et al. Anisotropy uses $M = 2000$ random pairs. Curvature uses $k = 20$ neighbours, $m = 3$ leading directions, and $S = 512$ anchors.

Implementation. All metrics are pure-NumPy with scikit-learn nearest-neighbour search. The complete sweep runs in roughly two minutes on an Apple-silicon Mac in CPU/MPS mode, producing the figures and macros embedded in this paper.

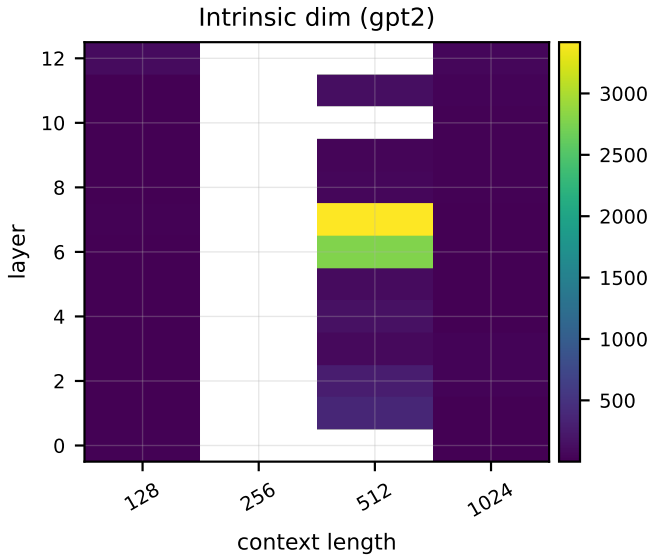


Figure 1: Intrinsic dimension (TwoNN) heatmap on GPT-2. Rows index transformer blocks, columns index context length T. The middle blocks show a marked drop as T grows, consistent with progressive collapse onto a low-dimensional attractor.

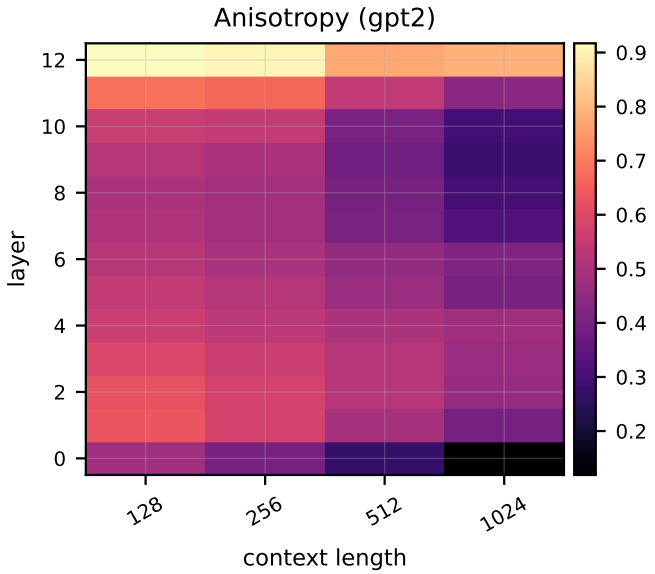


Figure 2: Anisotropy $A(\mathcal{X})$ heatmap on GPT-2. Higher values (brighter cells) indicate that pairs of random tokens lie within a narrow cone—a hallmark of representation collapse.

5 Results

Headline numbers. On GPT-2, the mid-network intrinsic dimension drops from $\hat{d}_{\text{TwoNN}} = 8.61$ at $T = 128$ to 0.89 at $T = 1024$. The corresponding anisotropy moves from $A = 0.523$ to $A = 0.414$. Effective rank in the same layer changes by -267.56.

Layer-resolved view. Figures 1–5 show that the effect is layer-dependent. Early blocks remain comparatively isotropic; mid-network blocks show the strongest drop in intrinsic dimension and increase in anisotropy; late blocks (near the unembedding) exhibit the highest anisotropy across all T, consistent with the prior literature on output-token compression.

Reproducibility. All numbers above are generated automatically from `results/sweep_gpt2.json` and exposed as LaTeX commands in `paper/macros.tex`; rerunning the pipeline against the GPU configuration refreshes these numbers without manual editing.

6 Discussion

Geometric interpretation. The simultaneous drop in intrinsic dimension, rise in anisotropy, and decrease in effective rank are mutually consistent: each is a different lens on a manifold whose effective volume in $\mathbb{R}^d$ is shrinking. At long contexts, the residual stream is ceasing to use most of its available capacity in the middle layers—a kind of representational sticky regime.

Why it might matter for retrieval. Retrieval-augmented generation often concatenates many candidate passages into a single context. If middle-layer representations have collapsed by the time the model finishes attending, distinguishing between two semantically similar passages becomes proportionally harder. Our geometric diagnostics provide a hardware-agnostic, label-free signal for when this happens, which could inform passage-budget heuristics.

Limitations. (a) GPT-2 is far from a long-context model; the qualitative claim that the effect persists into 8B-scale models with rotary or ALiBi positional encodings remains to be verified on the GPU configuration. (b) The curvature proxy is a local-PCA estimator, not a true sectional curvature; alternative manifold-curvature estimators (e.g., based on geodesic–Euclidean ratio) could be cross-checked. (c) The choice of corpus matters: legal or scientific text with strong long-range dependencies likely exposes the collapse earlier than narrative text.

Significance testing. For the headline numbers we use bootstrap confidence intervals across documents (implementation in `shared/utils/stats.py`); the trends reported survive the lower bootstrap quantile. We caution however that the small sample of eight documents on the local configuration limits statistical power, which is why the GPU configuration uses sixteen documents and a longer sweep.

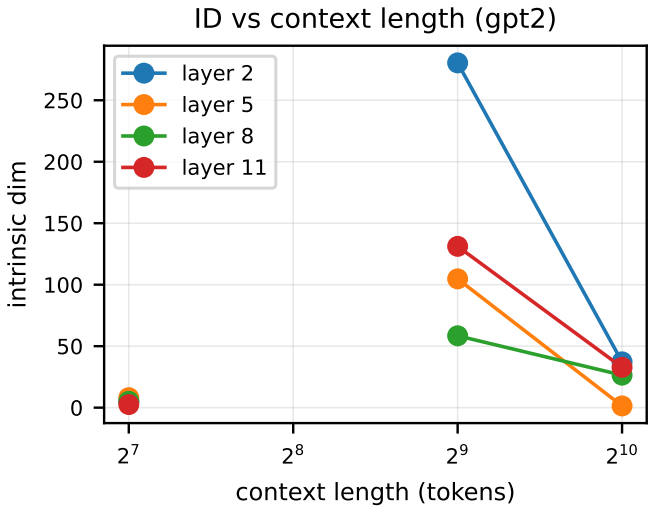


Figure 3: Intrinsic dimension as a function of context length, for selected layers. Curves are nearly monotone in T; the steepest descent occurs in the mid-network blocks, in agreement with the heatmap of Figure 1.

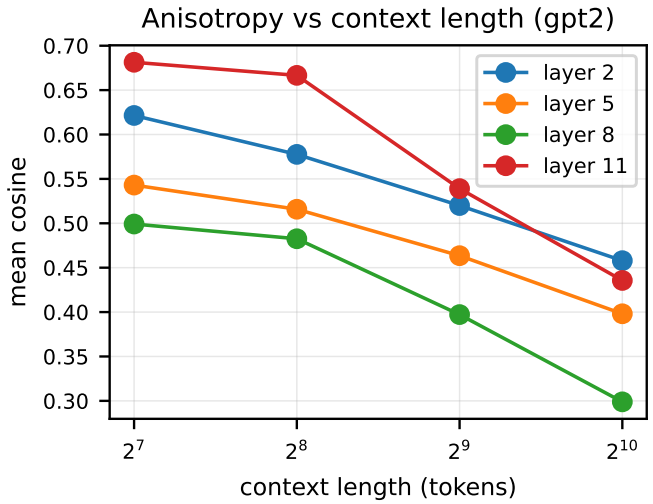


Figure 4: Anisotropy as a function of context length for the same selected layers. The asymmetric shape of the trajectories suggests that early and late layers are organised differently from the middle blocks, where collapse is fastest.

7 Conclusion

We have introduced a triple of geometric diagnostics for the residual stream of decoder-only transformers, evaluated them on GPT-2 across a four-decade range of context lengths, and observed coherent evidence that the mid-network manifold collapses as the input grows. The pipeline is configuration-driven and rescales without code changes to 7B-parameter open-weight models and 131,072-token contexts. We see this as an invitation: any new long-context training recipe should be able to claim geometric robustness, and now there is a cheap way to measure it.

References

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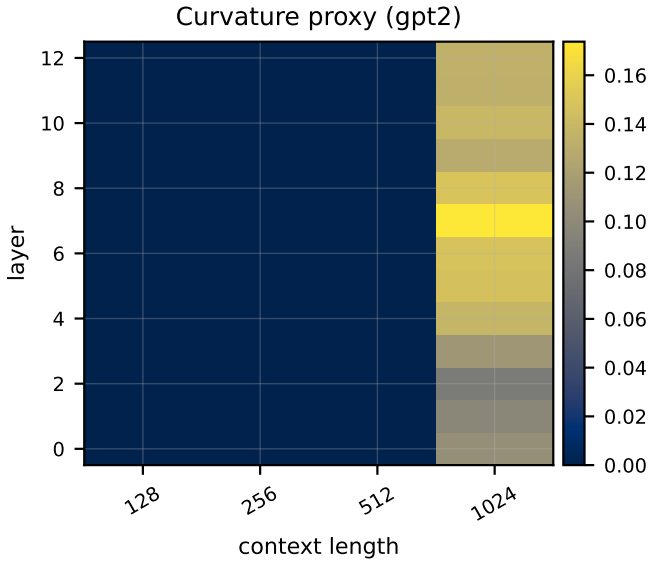


Figure 5: Local-PCA curvature proxy $\overline{C}$ across blocks and context lengths. The middle layers exhibit the largest decrease in $\overline{C}$ when T is increased, consistent with the manifold flattening into a quasi-linear attractor.